

International Transfers or Differentiated Carbon Prices?

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Abstract

The various princ

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Contents

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Introduction

1 Correspondence between transfers in a cap-and-trade and differentiated carbon prices

1.1 The static case

Setting

Decarbonization is costly. When a country i faces a carbon price p_i , it curtails consumption by investing in low-carbon equipment and reorienting consumption to less preferred, less carbon-intensive products. These costs to consumption are called abatement costs and denoted a_i . Taking into account the possibility of transfers t_i from the rest of the world, expressing everything in per capita terms, and abstracting from exogenous investments, country i 's consumption is related to its potential output y_i as follows:

$$c_i = y_i + t_i - a_i$$

Assume that the welfare of country i depends on its consumption c_i and global emissions $E = \sum_j e_j n_j$, where e_j is country's j per capita emissions and n_j its population size. For now, take global emissions as fixed at their optimal level E^* . Let p^* be the uniform carbon price required to attain E^* , i.e. $E^* = \sum_j e_j(p^*) n_j$. As shortcuts, we denote $e_i := e_i(p_i)$, $a_i := a_i(e_i)$, $e_i^* := e_i(p^*)$, and $a_i^* := a_i(e_i^*)$.

A situation is fully specified by the set of transfers $(t_j)_j$ and carbon prices $(p_j)_j$ for each country. For country i , given that global emissions are taken as fixed, the situation can be summarized as $s = (t_i, p_i)$, and (with a slight abuse of notation) we may express i 's welfare accordingly: $u_i(c_i; E^*) = u_i(t_i, p_i)$.

Equivalent variation from the benchmark in the absence of transfers

To analyze an arbitrary situation s , it will be useful to relate it to two special cases. First, the coordinated situation $c^* = (0, p^*)$ without transfer and with a uniform carbon price, which we take as a benchmark. Second, the autarky situation $a = (0, p_i)$, without transfer and with differentiated prices, which will be a useful intermediary to analyze an arbitrary situation. We denote $V_i(p_i)$ the equivalent variation between situation a and c^* , i.e. the transfer required at c^* to get a 's welfare:

$$\begin{aligned} u_i(V_i(p_i), p^*) &= u_i^a = u_i(0, p_i) \\ c_i^{c^*} + V_i(p_i) &= c_i^a \\ V_i(p_i) &= c_i^a - c_i^{c^*} \\ &= a_i(e_i^*) - a_i(e_i) \\ &\sim (e_i^* - e_i) \frac{da_i}{de}(e_i^*) = (e_i - e_i^*) p^* \end{aligned}$$

as $\frac{da_i}{de}(p) = -p$. Denoting $\varepsilon^* = -\frac{de_i/e_i^*}{dp/p^*}$ the absolute price elasticity of emissions at p^* , we have

$$V_i(p_i) \sim (p^* - p_i)e_i^*\varepsilon^*$$

V_i is positive iff $p^* > p_i$, as country i “needs” a positive compensation to get the same welfare as in a compared to c^* with a higher price p^* (and lower emissions).

Equivalent transfer for an arbitrary situation

Let g_i^s be country i 's net gain (or “equivalent transfer”) in a situation $s = (t_i, p_i)$ compared to the benchmark situation $c^* = (0, p^*)$:

$$\begin{aligned} g_i^s &= (c_i^s - c_i^a) + (c_i^a - c_i^{c^*}) \\ &= t_i + V_i(p_i) \end{aligned}$$

Note that we can call g_i^s the equivalent transfer as it is the transfer at zero price that makes i indifferent with s , i.e. the transfer such that $(g_i^s, 0)$ is welfare-equivalent to $s = (t_i, p_i)$. The equivalent transfer is a very useful metric to compare the welfare of a country in different situations.

Correspondence with an equivalent price

Now, we have all the tools to study the correspondence between transfers, differentiated emission rights, and differentiated prices. Let p_i^* be the “equivalent price” such that $s = (t_i, p_i)$ is welfare-equivalent to $s^* = (0, p_i^*)$.

$$\begin{aligned} g_i^s &= g_i^{s^*} \\ t_i + V_i(p_i) &= V_i(p_i^*) \\ t_i + (p^* - p_i)e_i^*\varepsilon^* &\sim (p^* - p_i^*)e_i^*\varepsilon^* \end{aligned}$$

Therefore, at the first order, the equivalent price is:

$$p_i^* \sim p_i - \frac{t_i}{e_i^*\varepsilon^*}$$

The equivalent price is negative when $p_i e_i^* \varepsilon^* < t_i$. As a negative price would require a transfer from the rest of the world, a regime with differentiated (positive) prices is restrictive compared to a system that allows for international transfers. Therefore, we prefer to use the concept of equivalent transfer g_i^s , rather than its dual the equivalent price.

Correspondence between differentiated prices and rights

Imagine that countries establish a uniform carbon price p^* , for example through a cap-and-trade. Say each person in country i has an emission right r_i , so they are entitled a claim $r_i p^*$ on global carbon revenue. Then, the net monetary transfer to i is

$$t_i^* = r_i p^* - e_i^* p^*$$

In the static case, there is a correspondence between emission rights r_i in a cap-and-trade and welfare-equivalent differentiated prices p_i in autarky (without transfers):

$$\begin{aligned} g_i^a &= V_i(p_i) = a_i^* - a_i \sim (e_i - e_i^*)p^* \sim (p^* - p_i)e_i^*\varepsilon^* \\ g_i^* &= t_i^* = (r_i - e_i^*)p^* \\ g_i^* = g_i^a &\Leftrightarrow t_i^* = a_i^* - a_i \\ &\Leftrightarrow (r_i - e_i^*)p^* \sim (p^* - p_i)e_i^*\varepsilon^* \end{aligned}$$

The left-hand side is the cap-and-trade equivalent transfer, which corresponds to the net monetary transfer received in the uniform cap-and-trade. This transfer is equal to the extra abatement cost incurred by going from autarky to the cap-and-trade without transfer. The (first-order) equivalence relation states that the transfer is negatively related to the difference between the autarky price and the global cap-and-trade price. The coefficient of this linear relation is the emissions times their price elasticity: the higher the emissions and the more they are sensitive to the carbon price, the larger the effect of the autarky carbon price on the cap-and-trade equivalent transfer.

One can easily isolate cap-and-trade emission rights or autarky price in function of the other variables:

$$\begin{aligned} g_i^* = g_i^a &\Leftrightarrow r_i \sim e_i \sim (1 + (1 - \frac{p_i}{p^*})\varepsilon^*)e_i^* \\ &\Leftrightarrow p_i \sim (1 + \frac{e_i^* - r_i}{e_i^*\varepsilon^*})p^* \end{aligned}$$

Efficiency of a uniform price

When there is undue inequality between countries, differentiated carbon prices can help improve the world's welfare, as reduced abatement in low-income countries act as a substitute for transfers. However, the Diamond-Mirrlees production efficiency theorem states that, in absence of other imperfections (such as market power or imperfect capital markets), direct transfers are always preferable to differentiated prices to address inequality. In our setting, this theorem means that carbon trading is always beneficial. More precisely, for any country i , compared to the autarky situation $(0, p_i)$, welfare is at least as high in the situation with uniform price where rights correspond to the autarky emissions: $(\tilde{t}_i = (e_i - e_i^*)p^*, p^*)$.

This results stems from the convexity of abatement costs, which gives:

$$a_i - a_i^* \geq (e_i - e^*) \frac{da_i^*}{de} = (e_i - e^*)(-p^*)$$

The transfer obtained from trading, $\tilde{t}_i$, is therefore greater than the transfer equivalent to autarky, t_i^* . Indeed:

$$\tilde{t}_i = (e_i - e_i^*)p^* \geq a_i^* - a_i = t_i^*$$

Rearranging the inequality and adding y_i , we can verify that the consumption is greater under trading: $\tilde{c}_i = y_i + \tilde{t}_i - a_i^* \geq y_i - a_i = c_i^a$, which proves the efficiency of trading at a uniform price.

1.2 The dynamic case

Setting

In this section, we will show that the correspondence between transfers and differentiated carbon prices imperfectly extends to the dynamic case.

Let us keep the previous notations and add an index t to denote the value of a variable at time t . Let β_t be the discount factor between the initial period and t (containing the pure rate of time preference and the welfare discounting from growth). Intertemporal values are denoted with uppercase letters. Monetary variables are discounted while physical ones are not, e.g. i 's intertemporal net gain is $G_i = \sum_t g_{it}\beta_t$ and its intertemporal emission right is $R_i = \sum_t r_{it}$.

A situation S is now given by the transfers and prices trajectories: $S = ((t_{it})_t, (p_{it})_t)$. Taking y_{it} as exogenous as a first approximation, for country i a situation S is welfare-equivalent to a situation S' with the same temperature trajectory iff:

$$\sum_t t_{it} - a_{it}(p_{it}) = \sum_t t'_{it} - a_{it}(p'_{it})$$

The uniform price without transfer case as benchmark

Consider the situation $C^* = (0, (p_t^*)_t)$ of a uniform carbon price trajectory $(p_t^*)_t$ and within-country revenue recycling (i.e. without transfers). The carbon price trajectory may be fixed in advance or emerge from the temporal allocation of the global intertemporal carbon budget (or emission right) R in a cap-and-trade system: $(r_t)_t$. Let us define the net gain for i at t in a situation S (with the same emissions trajectory as C^*) in relation to C^* . As above: $g_{it} = t_{it} + V_{it}(p_{it})$, where $V_{it}(p_{it}) = a_{it}^* - a_{it}$.

A more limited correspondence between differentiated prices and rights

As in the static case, let us look for a correspondence between an autarky situation A with differentiated prices and a cap-and-trade system U with uniform price. In the general case, $A = (0, (p_{it})_t)$, while $U = ((r_{it}p_t^* - e_{it}^*p_t^*)_t, (p_t^*)_t)$ is indirectly determined by $r = (r_{it})_t$. The welfare equivalence for i between A and U can be approximated as follows:

$$\begin{aligned} G_i^A &= \sum_t (a_{it}^* - a_{it})\beta_t \sim \sum_t (p_t^* - p_{it})e_{it}^*\varepsilon_t^*\beta_t \\ G_i^U &= \sum_t (r_{it} - e_{it}^*)p_t^*\beta_t \\ G_i^U &= G_i^A \Leftrightarrow \sum_t (r_{it} - e_{it}^*)\beta_t p_t^* \sim \sum_t (p_t^* - p_{it})e_{it}^*\varepsilon_t^*\beta_t \end{aligned}$$

Contrary to the static case, there is no direct equivalence between an intertemporal aggregate of the emission rights and an intertemporal aggregate of differentiated prices, since the welfare effect of emission rights depends on the cap-and-trade price trajectory p_t^* .

The same reason also complicates the comparison between two possible allocations r and r' of the intertemporal carbon budget in a cap-and-trade, even when the price trajectories coincide. Indeed, the condition $G_i^U > G_i^{U'} \Leftrightarrow \sum_t r_{it} \beta_t p_t^* > \sum_t r'_{it} \beta_t p_t^*$ does *not* imply that $R_i = \sum_t r_{it} > \sum_t r'_{it} = R'_i$. In other words, **there is no immediate relation between i 's intertemporal carbon budget and its intertemporal welfare, as the price trajectory matters.**

Let us study whether we can find a correspondence in simpler cases. For the autarky situation A , consider a homothetic carbon price vis-à-vis the world average: $p_{it} = \pi_i p_t$. For the cap-and-trade situation U , consider that i is granted a fixed share of the world's carbon budget: $r_{it} = r_i r_t$. Now, the equivalence between A and U rewrites:

$$G_i^U = G_i^A \Leftrightarrow r_i \sum_t r_t \beta_t p_t^* - \sum_t e_{it}^* p_t^* \beta_t \sim \sum_t (p_t^* - \pi_i p_t) e_{it}^* \epsilon_t^* \beta_t$$

Provided that the price trajectories in autarky are homothetic to the price trajectory in the cap-and-trade (without loss of generality): $p_{it} = \pi_i p_t^*$; the equivalence simplifies further:

$$G_i^U = G_i^A \Leftrightarrow r_i \mathcal{B} \sim \mathcal{E}_i^* + (1 - \pi_i) \Delta \mathcal{A}$$

where $\mathcal{B} = \sum_t r_t \beta_t p_t^*$, $\mathcal{E}_i^* = \sum_t e_{it}^* p_t^* \beta_t$, and $\Delta \mathcal{A} = \sum_t p_t^* e_{it}^* \epsilon_t^* \beta_t$.

Although we do find a correspondence in this special case, the formula faces practical limitations, as **it requires knowing the future price trajectory $(p_t^*)_t$, which can only be imprecisely estimated** in the cap-and-trade situation. In any case, although one can find a trajectory of differentiated prices welfare-equivalent to a given allocation of emissions rights, and vice versa, the derivation cannot be done analytically, **one has to resort to an integrated assessment model.**

However, **in the Hotelling case** (with a fixed carbon budget, perfect foresight, and efficient capital markets), the formula simplify further. Indeed, the Hotelling price grows at the discount rate, so that the discounted price is constant: $\beta_t p_t^* = p_0$. Only in this very special case, **the formula simplifies to a formula analogous to the static case formula:**

$$G_i^U = G_i^A \Leftrightarrow R_i - E_i^* \sim (1 - \pi_i) \sum_t e_{it}^* \epsilon_t^*$$

2 Differentiated carbon price floors

Some authors propose that all countries in a coalition price carbon nationally, without cross-country revenue sharing but with differentiated carbon price floors based on income levels¹ (??). In a prominent contribution, ? propose restricting the agreement to carbon-intensive manufacturing products (e.g. steel, aluminum, cement) and applying a carbon border adjustment mechanism (CBAM) only at the borders of the climate coalition

¹Namely, the authors propose \$75/tCO₂ for high-income countries, \$50/t for upper middle-income countries, and \$25/t for lower-income countries.

(at \$75/tCO₂). This proposal would have the EU renounce its own CBAM in exchange for China pricing carbon emissions in its manufacturing sector — thereby increasing the price of its final products (?). This proposal would reduce global CO₂ emissions by 2%. It has little chances of being implemented, as China represents about 70% of emissions covered in the proposed coalition, and China appears unwilling to commit to the proposed carbon price level (\$50/t).

One may wonder why carbon prices should be differentiated across countries. From a theoretical perspective, absent any imperfections, a uniform carbon price coupled with cross-country transfers is optimal (?). Specifically, a uniform price is more efficient, and under a redistributive system, lower-income countries would gain purchasing power. Indeed, provided lower-income countries are allocated more emission rights than actual needs, they could in principle choose to keep their emissions stable while receiving a financial transfer. Yet, the high carbon price would provide incentives to decarbonize, allowing them to benefit from larger transfers.

Admittedly, four kinds of imperfections justify differentiated carbon prices across countries or sectors: divergent discount rates, the presence of country-specific distortive taxes, market power in trade, and constraints preventing cross-country transfers. Let us review them in turn. First, ? show that equalizing carbon prices across countries is inefficient “if the equilibrium features cross-country differences in discount rates (or interest rates)”, themselves due to inefficiencies in the allocation of capital.² However, no model has been developed to know whether welfare would increase or decrease upon equalizing carbon prices (through emission trading). Second, ? explain theoretically and ? models numerically that existing distortions can be reduced by differentiating carbon prices across sectors or countries. For example, if aviation remains under-taxed, then a higher sector-specific carbon price improves welfare. Third, the terms-of-trade effect can be well understood taking the example of China, which possesses significant market power in the oil or manufactured goods markets. As an oil importer, China would benefit from a lower oil price. By taxing oil more than other sectors, China would reduce global demand for oil, thereby lowering its price and improving its terms-of-trade. Similarly, China has an interest in setting a higher carbon price on manufactured goods to increase the value of its exports. Fourth, differentiated prices are generally justified by the assertion that international transfers are politically infeasible (??). ? refer to an “efficiency-sovereignty” trade-off in climate policies: either carbon prices are differentiated and efficiency is lost, or a uniform price is applied alongside transfers to equalize efforts across countries. Because they deem transfers infeasible, they view differentiated prices as a necessary second-best solution.

However, there is a distributional equivalence between differentiated carbon prices and a uniform price with differentiated emission rights (?). More precisely, a uniform price combined with an appropriate allocation of emission rights can replicate the distribution of costs and benefits generated by any differentiated price-floor agreement, while

²While differentiated carbon prices might be optimal in a second-best world, the first best is in principle attainable through capital market reforms to correct their imperfections.

153 typically delivering the same or greater global emission reductions. This observation
154 should prompt governments to question their apparent reluctance to implement trans-
155 fers, given that they produce the same distributive effects as differentiated prices without
156 associated inefficiency costs.

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Supplementary Tables

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Beyond Current Initiatives:

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ITMO Regulations to Ratchet up Ambition